

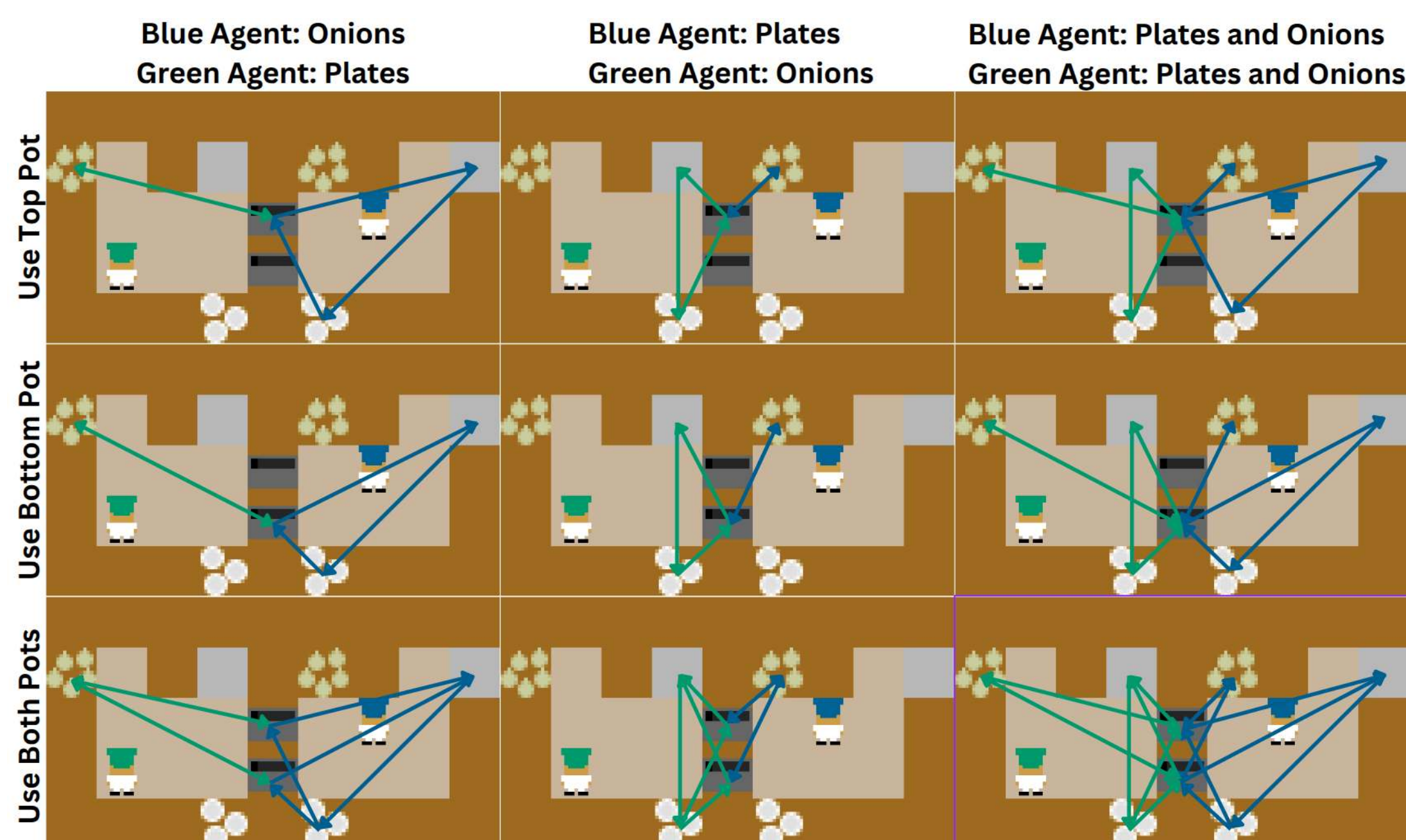


Traditional Reinforcement Learning **Struggles** to **Adapt** to Implicit Human Strategies

- Autonomous agents in human environments must be able to collaborate without relying on direct language instruction.
- Existing RL agents optimize only the task reward, and do not adapt well to changing strategies nor offer interpretability.
- Overcooked-AI environment [1] used as a testbed to study how robots can infer implicit intentions from human behavior.

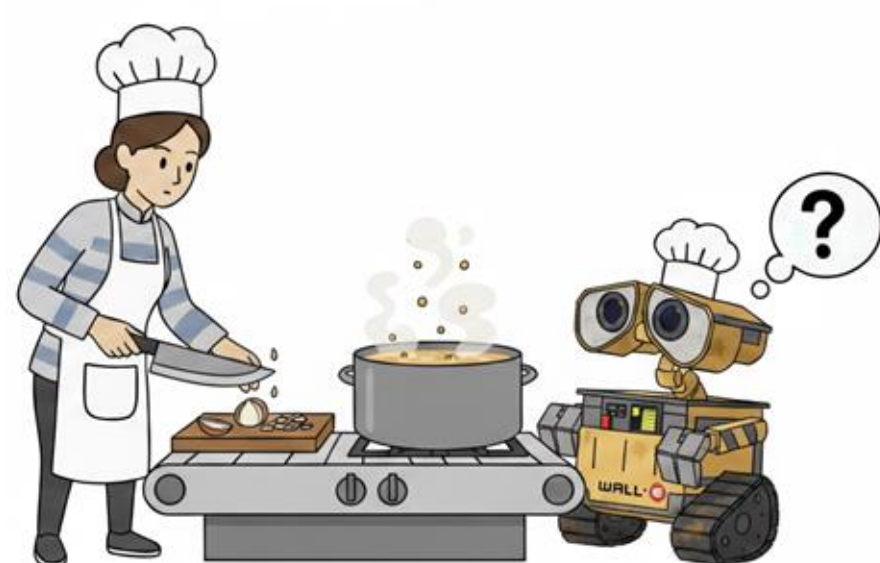
Dataset Generated Using Asymmetric Layout Enables **Strategic Variation**

- 480 human-human trajectory pairs of 60 seconds each.

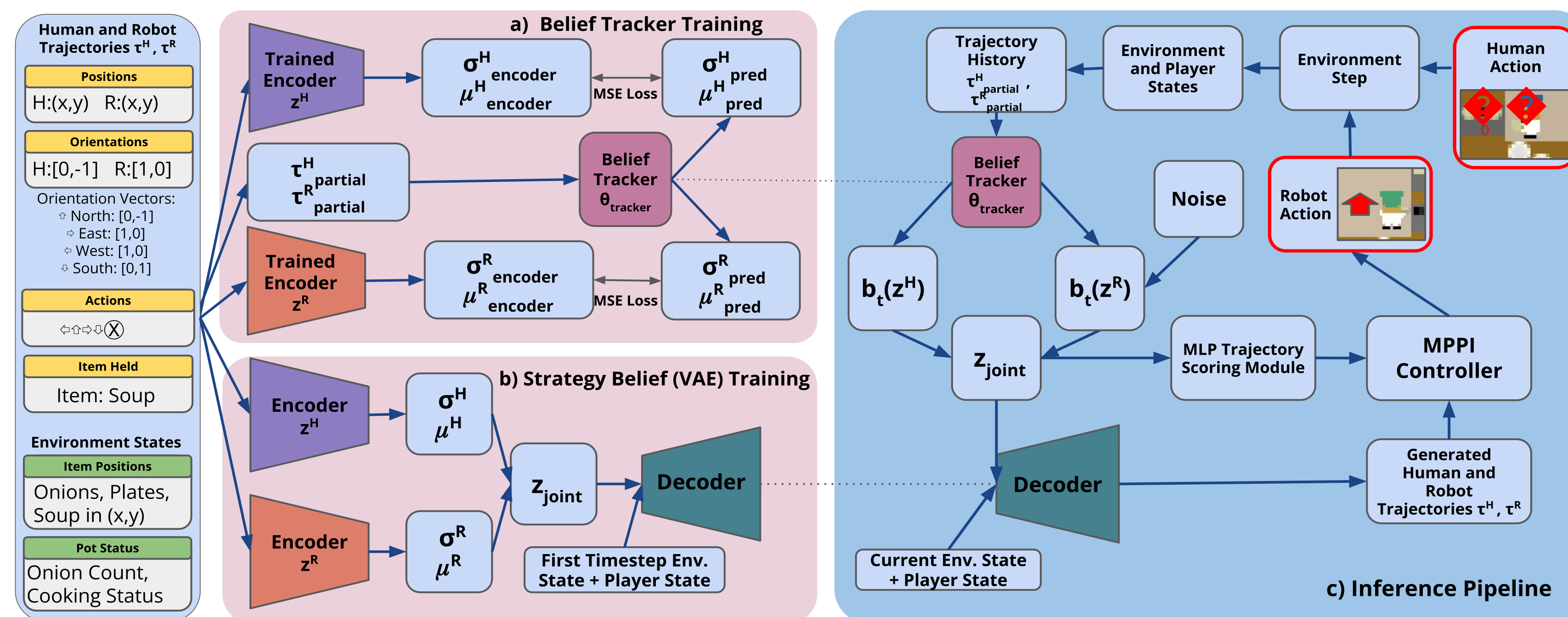


Human-Robot Teaming is **Modeled** as a **Latent-MDP** to Capture Unobservable Intent

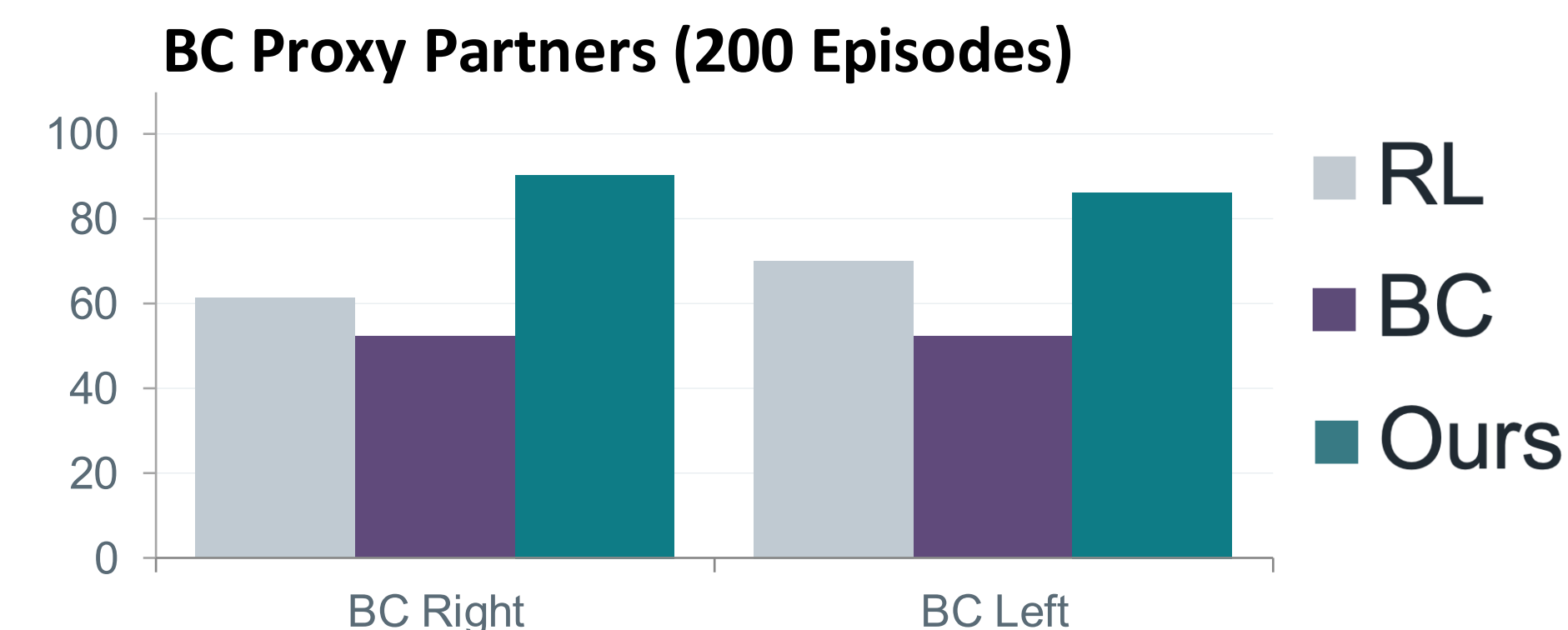
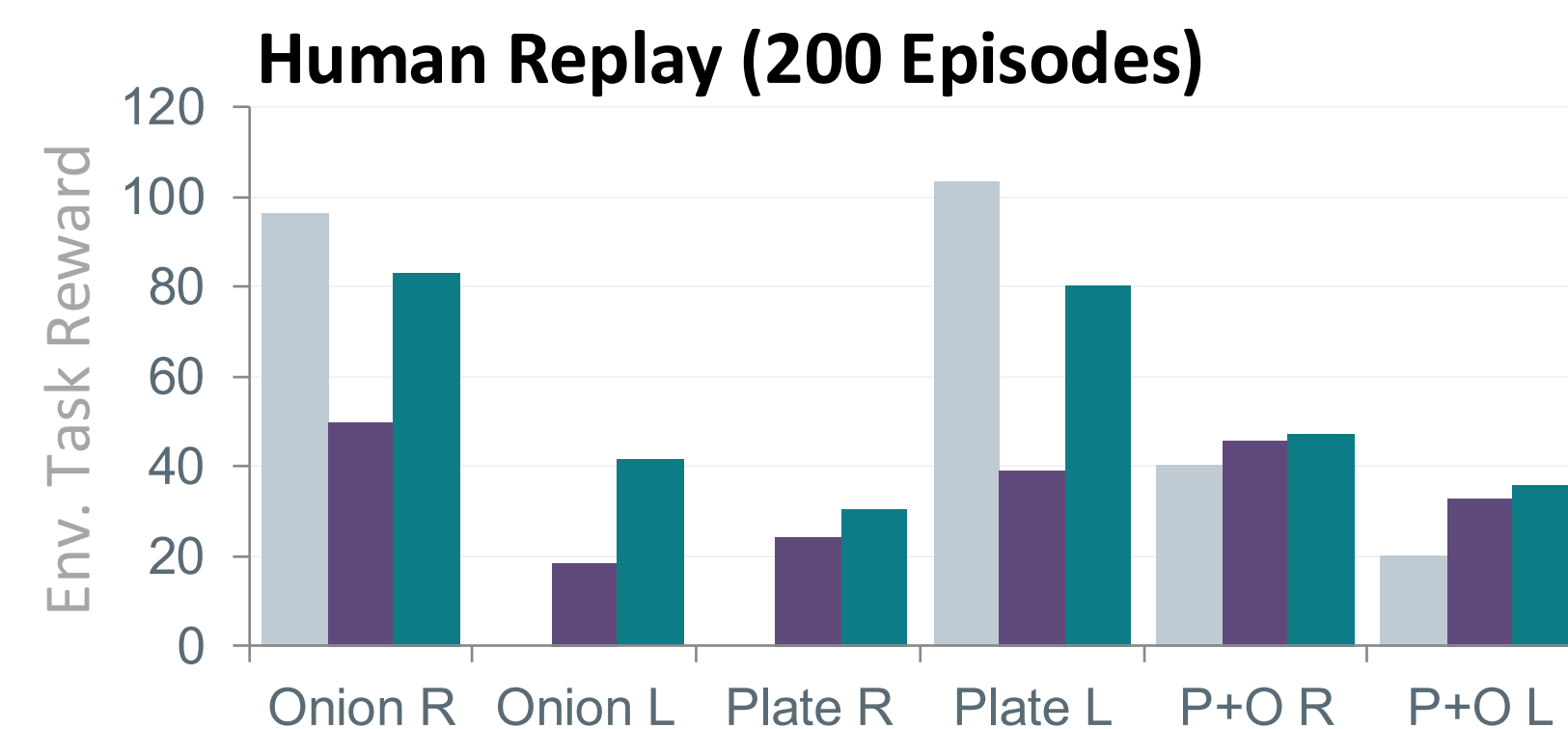
- POMDP is reduced to a latent MDP by modeling strategy as a low dimensional latent variable.
- Partial observability comes from the human partner's underlying policy being unknown to the agent.



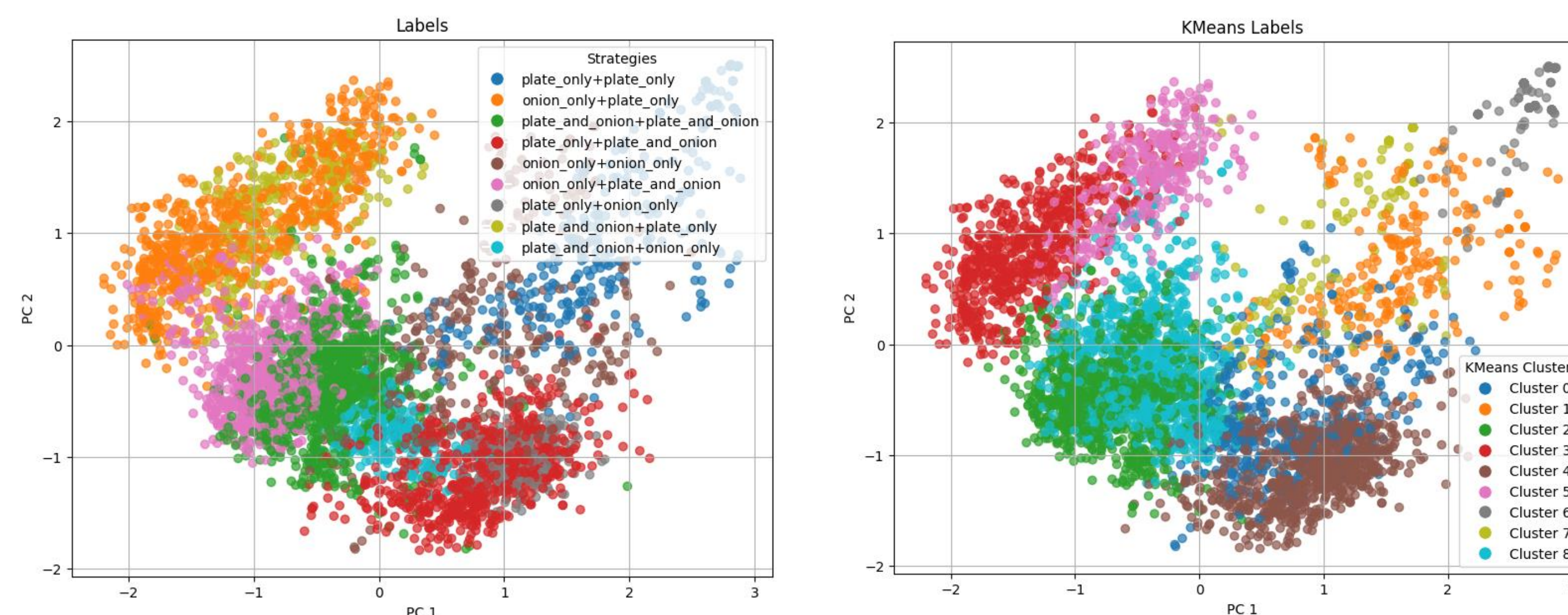
A Joint VAE and MPPI Controller **Bridge** High-Level Strategy with Low-Level Actions



VAE + MPPI Outperform RL and BC Baselines in Task Performance



VAE Successfully Grouped Strategies by their **Functional Similarity**



[1] Carroll et al. (2019), "On the utility of learning about humans for human-ai coordination", NeurIPS